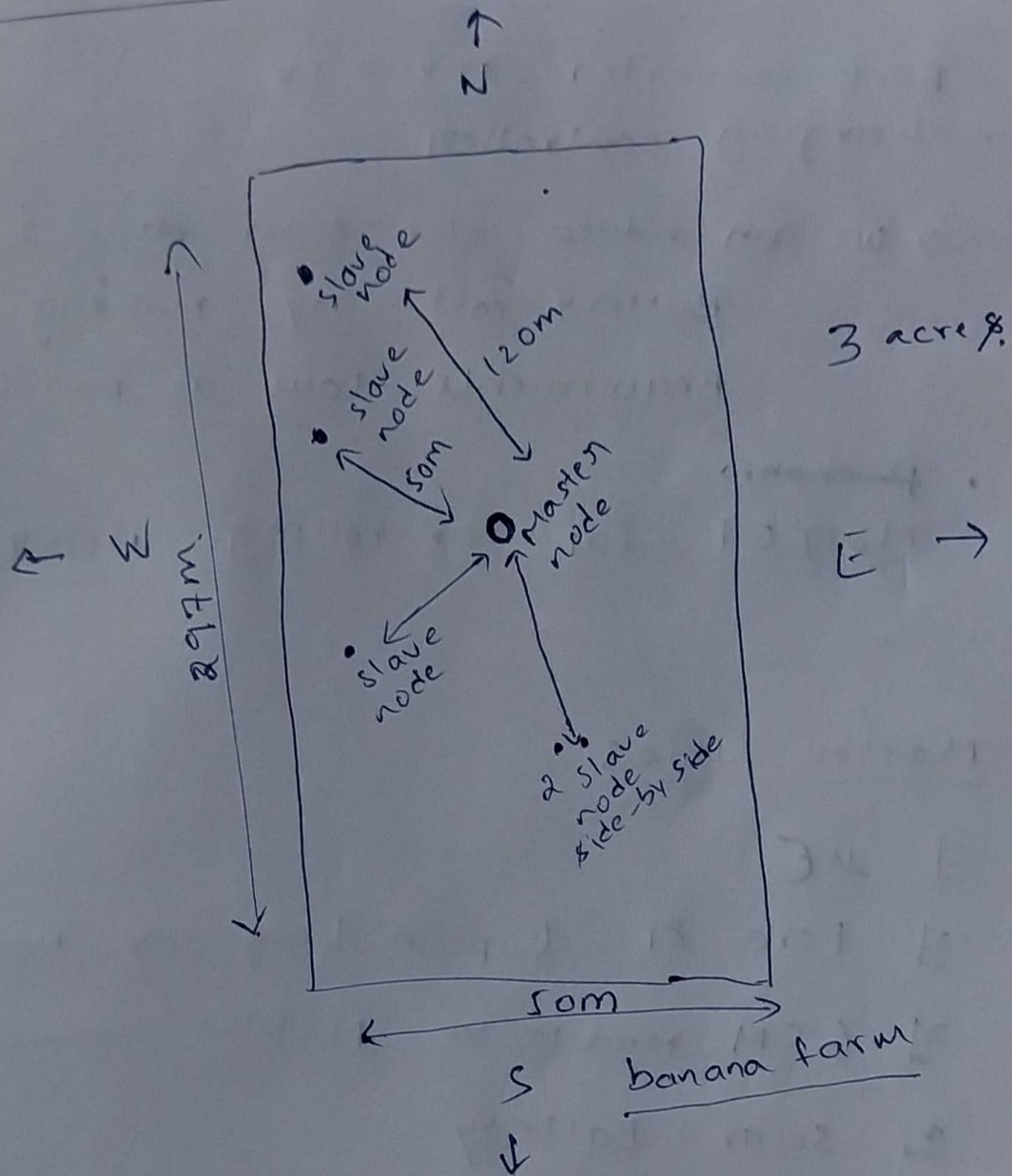


Control Mechanism Remotely.



Agenda

- Just to control valve open / close
 - Use less power to make budget
 - Make the setup to ^{friendly} work at least 3 years
- slave node components

- 1) MC [ESP32 / NodeMCU]
- 2) LORA or ESP in Long Range Mode.
- 3) Motor driven. [BT57960]
- 4) ^{12V} DC gear motor 10-30 RPM., 20-25kgcm
- 5) Limit switches for status of valve.
- 6) Battery + solar panel [Based on power requirement]

- Buck converter 12V to 5V
- charging controller.

→ We can reduce the solar panel & Battery cost by making power Requirements low. as possible.

- ~~p channel~~ MosFET for controlling motor

Master Node:

- 1] MC
- 2] Lora or Esp in Long range mode
- 3] GSM module to Mobile communication
- 4] Solar + Battery.

Slave node cost

- MC - ₹150 - 200
- Lora, with antenna - ₹200 - 250
- Driver - ₹200
- Dc motor - ₹400
- Switches - $2 \times 20 = ₹40$
- Battery - 12V Amps depend on power required

under ₹500,

Student ID	Student Name

solar - 6W under £500
along with charge controller

Buck converter - under £100.

MOSFET - £10.

Master Node cost.

→ MC - £150-200

→ LoRa £200-250

→ GSM £500-700

→ solar + battery. based on power requirement.

challenges

power requirement = money

We can cut solar panel from
project if management of
long battery life

farmer just sends command
valve - ON/OFF

on - an average actuator needs
power for 10-20 seconds.

→ Making node ultra low consuming
power saves budget.

My thoughts on power reducing

→ Using RTC setting the schedule during which farmers message can be send to nodes

Remaining time they are sleep state cut off power to all other modules using MOSFET.

→ continuous power is required is only for

1) controller

2) Lora

3) Driver

Management

1) controller → usage of deep sleep

2) Lora & Driver → using MOSFET we can cut power.

As Actuator need power for less time we don't need 2nd thought as we are using Gear DC motor it doesn't use the continuous power.